

# A Catalog of Ghosts: the 120 Pilot Candidate Counterexamples and Sturmian Realizability Data

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## Abstract

This catalog accompanies the Shadow Frontier project (successor to *Finite Spectral Shadows for the Collatz Valuation Cocycle*). It lists the 120 candidate-counterexample valuation words of pilot run 001 (deterministic regeneration, seed 1): words of length  $M = 240$  over the alphabet  $\{1, 2, 4\}$ , calibrated to mean  $\log_2 3$  within  $2/M$  and carrying a non-coboundary order-3 shadow bias, sampled from the law  $p_1 = 0.567$ ,  $p_2 = 0.357$ ,  $p_4 = 0.076$ . Every one of the 120 is a *mixed-adic ghost*: its 2-adic realizability class modulo  $2^{B_M+1}$  has least positive representative of nearly full bit length, not stabilizing between prefix lengths  $M/2$  and  $M$  — the signature that no positive integer realizes the word. We also list ten Sturmian ghosts of the critical slope  $\alpha = 2 - \log_2 3$  (all Sturmian words are now *provably* non-realizable, by the repetition-rigidity theorem); their data is shown for comparison. The objects below are points of  $\mathbb{Z}_2$ ; the Collatz conjecture asserts that classes like these never meet  $\mathbb{Z}^+$ .

## 1 Conventions

For a valuation word  $w = (b_0, \dots, b_{M-1})$ ,  $B_N = \sum_{n < N} b_n$ ; the set of odd  $x$  whose first  $N$  valuations equal the prefix is a single residue class modulo  $2^{B_N+1}$ ; “rep bits” is the bit length of its least positive representative  $r_N$ , “mod bits” is  $B_N + 1$ , and the fill ratio is their quotient. A positive realization would force  $r_N$  to stabilize at  $\lceil \log_2 x_0 \rceil$  once  $2^{B_N} > x_0$ ; a ghost rides the diagonal  $r_N \sim 2^{B_N}$ . The bias is  $|D_3| = \left| \frac{1}{M} \sum_n \zeta_3^{-b_n} - \mathbb{E}_{\text{geom}}[\zeta_3^{-b}] \right|$ , the order-3 non-coboundary two-point shadow.

## 2 Translating between a word and the integer it represents

### Explanation

A word does not represent a single integer but a single *residue class*: the set of odd  $x$  whose first  $M$  valuations equal  $w = (b_0, \dots, b_{M-1})$  is exactly one arithmetic progression

$$x \equiv r \pmod{2^{B_M+1}}, \quad B_M = \sum_{n < M} b_n,$$

and “the integer” tabulated in this catalog is its least positive representative  $r$ . Every member  $r + k 2^{B_M+1}$  ( $k \geq 0$ ) produces the same first  $M$  letters; the word sees only  $x \bmod 2^{B_M+1}$ . The converse direction is trivial: iterate  $x \mapsto (3x + 1)/2^{v_2(3x+1)}$  and read the exponents.

### Formula

Along the prefix the orbit point has the exact affine form

$$x_n = \frac{3^n x_0 + A_n}{2^{B_n}}, \quad A_n = \sum_{j < n} 3^{n-1-j} 2^{B_j}, \quad B_n = \sum_{j < n} b_j.$$

The letter condition  $v_2(3x_n + 1) = b_n$  is the congruence

$$3^{n+1}x_0 + A_{n+1} \equiv 2^{B_{n+1}} \pmod{2^{B_{n+1}+1}} \iff x_0 \equiv 3^{-(n+1)}(2^{B_{n+1}} - A_{n+1}) \pmod{2^{B_{n+1}+1}},$$

(3 is a unit in  $\mathbb{Z}_2$ ), so each letter appends  $b_n$  new binary digits of  $x_0$ : a Hensel-style lift. After  $M$  letters the class modulus is  $2^{B_M+1}$  and  $r$  is the lift's value reduced to  $[1, 2^{B_M+1})$ .

## Python

The lift as implemented in `search/frontier.py` (`realize`);  $r$  is built letter by letter:

```
def realize(bs):
    # word -> (r, B+1)
    r, Mb = 1, 1
    # class rep and its bit-modulus
    A = 0; B = 0; P3 = 1
    # running A_n, B_n, 3^n
    for b in bs:
        need = B + b; mod = 1 << (need + 1)
        c = (3*A + (1 << B)) % mod
        # A_{n+1} mod 2^{B+b+1}
        x0 = (pow(3*P3, -1, mod)
              * (((1 << need) - c) % mod)) % mod
        if need + 1 > Mb:
            r = x0; Mb = need + 1
        A = 3*A + (1 << B); P3 *= 3; B += b
    return r, Mb

def word_of(x, M):
    # integer -> word
    out = []
    for _ in range(M):
        t = 3*x + 1; b = (t & -t).bit_length() - 1
        out.append(b); x = t >> b
    return out
```

## Commands

```
cd shadow_frontier
# word -> integer (paste the letters as one string)
python3 -c "import sys; sys.path.insert(0,'search'); \
from frontier import realize; \
w=[int(c) for c in '111411211121...']; \
r,Mb=realize(w); print(r, r.bit_length(), Mb)"
# integer -> word, plus the four-panel candidate image
python3 search/candidate_image.py --word 1,1,1,4,1,1,2,... --out viz/cand.png
python3 search/candidate_image.py --seed 687871 --steps 55 --out viz/c687871.png
```

## Worked example: catalog word #1

For the word #1 above ( $M = 240$ , mean 1.583,  $|D_3| = 0.214$ ):  $B_M = 380$ , modulus  $2^{381}$ , and the least positive representative is the 381-bit integer

45276335476484047115725251008950986904354815513999907611  
28044450220436219657540681487193488995285987515743750727311

whose first 240 Syracuse letters reproduce the word exactly (machine-checked). Its fill ratio is  $r$ 's bit length over  $B_M + 1$ :  $381/381 = 1.000$  — and the half-word prefix already needs 184 bits of 185: the representative *grows with the prefix* (184  $\rightarrow$  381 bits from  $M = 120$  to 240) instead of stabilizing. That diagonal ride is the ghost signature of the Conventions section: the class contains no positive integer below  $2^{380}$ . Contrast a true integer realization: the survival champion 687871 *is* the least representative of its own 50-letter class, and stays put as the prefix grows — stabilization versus diagonal is the whole test.

### 3 The 120 pilot candidates: statistics

All 120 are ghosts: no representative stabilized; fill ratios  $\approx 1$ .

| #  | mean $b$ | $ D_3 $ | $P(16)$ | $B_{240}$ | rep bits | mod bits | fill  |
|----|----------|---------|---------|-----------|----------|----------|-------|
| 1  | 1.5833   | 0.214   | 225     | 380       | 381      | 381      | 1.000 |
| 2  | 1.5792   | 0.214   | 225     | 379       | 379      | 380      | 0.997 |
| 3  | 1.5792   | 0.214   | 225     | 379       | 378      | 380      | 0.995 |
| 4  | 1.5833   | 0.239   | 205     | 380       | 381      | 381      | 1.000 |
| 5  | 1.5917   | 0.220   | 213     | 382       | 383      | 383      | 1.000 |
| 6  | 1.5792   | 0.215   | 225     | 379       | 379      | 380      | 0.997 |
| 7  | 1.5917   | 0.215   | 225     | 382       | 383      | 383      | 1.000 |
| 8  | 1.5792   | 0.216   | 225     | 379       | 380      | 380      | 1.000 |
| 9  | 1.5792   | 0.229   | 214     | 379       | 380      | 380      | 1.000 |
| 10 | 1.5792   | 0.233   | 225     | 379       | 380      | 380      | 1.000 |
| 11 | 1.5792   | 0.229   | 225     | 379       | 379      | 380      | 0.997 |
| 12 | 1.5792   | 0.216   | 225     | 379       | 380      | 380      | 1.000 |
| 13 | 1.5917   | 0.233   | 193     | 382       | 382      | 383      | 0.997 |
| 14 | 1.5792   | 0.215   | 224     | 379       | 380      | 380      | 1.000 |
| 15 | 1.5917   | 0.216   | 208     | 382       | 383      | 383      | 1.000 |
| 16 | 1.5833   | 0.214   | 225     | 380       | 378      | 381      | 0.992 |
| 17 | 1.5917   | 0.215   | 225     | 382       | 380      | 383      | 0.992 |
| 18 | 1.5792   | 0.218   | 225     | 379       | 380      | 380      | 1.000 |
| 19 | 1.5917   | 0.220   | 225     | 382       | 383      | 383      | 1.000 |
| 20 | 1.5792   | 0.260   | 214     | 379       | 380      | 380      | 1.000 |
| 21 | 1.5792   | 0.272   | 208     | 379       | 380      | 380      | 1.000 |
| 22 | 1.5917   | 0.220   | 214     | 382       | 383      | 383      | 1.000 |
| 23 | 1.5875   | 0.228   | 225     | 381       | 382      | 382      | 1.000 |
| 24 | 1.5875   | 0.246   | 214     | 381       | 380      | 382      | 0.995 |
| 25 | 1.5917   | 0.222   | 211     | 382       | 383      | 383      | 1.000 |
| 26 | 1.5917   | 0.242   | 189     | 382       | 383      | 383      | 1.000 |
| 27 | 1.5792   | 0.222   | 222     | 379       | 376      | 380      | 0.989 |
| 28 | 1.5917   | 0.214   | 211     | 382       | 383      | 383      | 1.000 |
| 29 | 1.5875   | 0.215   | 225     | 381       | 382      | 382      | 1.000 |
| 30 | 1.5792   | 0.260   | 217     | 379       | 375      | 380      | 0.987 |
| 31 | 1.5917   | 0.214   | 216     | 382       | 382      | 383      | 0.997 |
| 32 | 1.5833   | 0.239   | 203     | 380       | 380      | 381      | 0.997 |
| 33 | 1.5792   | 0.218   | 225     | 379       | 377      | 380      | 0.992 |
| 34 | 1.5875   | 0.217   | 225     | 381       | 380      | 382      | 0.995 |
| 35 | 1.5917   | 0.214   | 224     | 382       | 379      | 383      | 0.990 |
| 36 | 1.5917   | 0.216   | 225     | 382       | 377      | 383      | 0.984 |
| 37 | 1.5792   | 0.220   | 225     | 379       | 380      | 380      | 1.000 |
| 38 | 1.5917   | 0.225   | 225     | 382       | 383      | 383      | 1.000 |
| 39 | 1.5875   | 0.221   | 225     | 381       | 381      | 382      | 0.997 |
| 40 | 1.5792   | 0.218   | 225     | 379       | 379      | 380      | 0.997 |
| 41 | 1.5917   | 0.225   | 206     | 382       | 380      | 383      | 0.992 |
| 42 | 1.5792   | 0.238   | 212     | 379       | 377      | 380      | 0.992 |
| 43 | 1.5792   | 0.215   | 225     | 379       | 375      | 380      | 0.987 |
| 44 | 1.5917   | 0.215   | 225     | 382       | 383      | 383      | 1.000 |
| 45 | 1.5875   | 0.217   | 225     | 381       | 379      | 382      | 0.992 |
| 46 | 1.5792   | 0.215   | 219     | 379       | 378      | 380      | 0.995 |
| 47 | 1.5792   | 0.229   | 223     | 379       | 380      | 380      | 1.000 |
| 48 | 1.5792   | 0.215   | 225     | 379       | 377      | 380      | 0.992 |
| 49 | 1.5917   | 0.215   | 225     | 382       | 383      | 383      | 1.000 |
| 50 | 1.5917   | 0.216   | 225     | 382       | 382      | 383      | 0.997 |

| #   | mean $b$ | $ D_3 $ | $P(16)$ | $B_{240}$ | rep bits | mod bits | fill  |
|-----|----------|---------|---------|-----------|----------|----------|-------|
| 51  | 1.5792   | 0.229   | 223     | 379       | 377      | 380      | 0.992 |
| 52  | 1.5792   | 0.248   | 206     | 379       | 380      | 380      | 1.000 |
| 53  | 1.5917   | 0.218   | 224     | 382       | 376      | 383      | 0.982 |
| 54  | 1.5792   | 0.215   | 225     | 379       | 380      | 380      | 1.000 |
| 55  | 1.5917   | 0.233   | 225     | 382       | 382      | 383      | 0.997 |
| 56  | 1.5917   | 0.279   | 194     | 382       | 383      | 383      | 1.000 |
| 57  | 1.5792   | 0.215   | 225     | 379       | 380      | 380      | 1.000 |
| 58  | 1.5792   | 0.218   | 225     | 379       | 377      | 380      | 0.992 |
| 59  | 1.5792   | 0.238   | 201     | 379       | 380      | 380      | 1.000 |
| 60  | 1.5792   | 0.214   | 225     | 379       | 380      | 380      | 1.000 |
| 61  | 1.5792   | 0.238   | 225     | 379       | 380      | 380      | 1.000 |
| 62  | 1.5833   | 0.230   | 213     | 380       | 381      | 381      | 1.000 |
| 63  | 1.5792   | 0.222   | 219     | 379       | 379      | 380      | 0.997 |
| 64  | 1.5792   | 0.215   | 219     | 379       | 379      | 380      | 0.997 |
| 65  | 1.5792   | 0.222   | 225     | 379       | 380      | 380      | 1.000 |
| 66  | 1.5917   | 0.216   | 196     | 382       | 383      | 383      | 1.000 |
| 67  | 1.5917   | 0.215   | 225     | 382       | 383      | 383      | 1.000 |
| 68  | 1.5792   | 0.216   | 225     | 379       | 372      | 380      | 0.979 |
| 69  | 1.5917   | 0.216   | 225     | 382       | 382      | 383      | 0.997 |
| 70  | 1.5917   | 0.216   | 225     | 382       | 383      | 383      | 1.000 |
| 71  | 1.5917   | 0.214   | 224     | 382       | 382      | 383      | 0.997 |
| 72  | 1.5792   | 0.214   | 225     | 379       | 380      | 380      | 1.000 |
| 73  | 1.5875   | 0.221   | 225     | 381       | 379      | 382      | 0.992 |
| 74  | 1.5917   | 0.215   | 222     | 382       | 383      | 383      | 1.000 |
| 75  | 1.5792   | 0.215   | 225     | 379       | 380      | 380      | 1.000 |
| 76  | 1.5792   | 0.225   | 225     | 379       | 380      | 380      | 1.000 |
| 77  | 1.5833   | 0.241   | 225     | 380       | 381      | 381      | 1.000 |
| 78  | 1.5917   | 0.215   | 225     | 382       | 382      | 383      | 0.997 |
| 79  | 1.5875   | 0.221   | 225     | 381       | 380      | 382      | 0.995 |
| 80  | 1.5875   | 0.217   | 225     | 381       | 381      | 382      | 0.997 |
| 81  | 1.5917   | 0.225   | 220     | 382       | 380      | 383      | 0.992 |
| 82  | 1.5833   | 0.214   | 222     | 380       | 381      | 381      | 1.000 |
| 83  | 1.5917   | 0.215   | 225     | 382       | 381      | 383      | 0.995 |
| 84  | 1.5917   | 0.218   | 225     | 382       | 383      | 383      | 1.000 |
| 85  | 1.5792   | 0.222   | 225     | 379       | 378      | 380      | 0.995 |
| 86  | 1.5792   | 0.238   | 225     | 379       | 380      | 380      | 1.000 |
| 87  | 1.5792   | 0.229   | 225     | 379       | 376      | 380      | 0.989 |
| 88  | 1.5875   | 0.227   | 225     | 381       | 382      | 382      | 1.000 |
| 89  | 1.5792   | 0.218   | 225     | 379       | 380      | 380      | 1.000 |
| 90  | 1.5917   | 0.216   | 225     | 382       | 383      | 383      | 1.000 |
| 91  | 1.5792   | 0.216   | 225     | 379       | 378      | 380      | 0.995 |
| 92  | 1.5833   | 0.225   | 225     | 380       | 381      | 381      | 1.000 |
| 93  | 1.5917   | 0.220   | 220     | 382       | 382      | 383      | 0.997 |
| 94  | 1.5792   | 0.214   | 225     | 379       | 380      | 380      | 1.000 |
| 95  | 1.5792   | 0.238   | 223     | 379       | 380      | 380      | 1.000 |
| 96  | 1.5792   | 0.238   | 225     | 379       | 379      | 380      | 0.997 |
| 97  | 1.5917   | 0.233   | 221     | 382       | 383      | 383      | 1.000 |
| 98  | 1.5792   | 0.229   | 219     | 379       | 380      | 380      | 1.000 |
| 99  | 1.5917   | 0.218   | 223     | 382       | 382      | 383      | 0.997 |
| 100 | 1.5792   | 0.215   | 225     | 379       | 379      | 380      | 0.997 |
| 101 | 1.5792   | 0.218   | 225     | 379       | 380      | 380      | 1.000 |
| 102 | 1.5792   | 0.218   | 216     | 379       | 376      | 380      | 0.989 |
| 103 | 1.5792   | 0.229   | 225     | 379       | 379      | 380      | 0.997 |
| 104 | 1.5875   | 0.217   | 225     | 381       | 382      | 382      | 1.000 |

| #   | mean $b$ | $ D_3 $ | $P(16)$ | $B_{240}$ | rep bits | mod bits | fill  |
|-----|----------|---------|---------|-----------|----------|----------|-------|
| 105 | 1.5917   | 0.229   | 211     | 382       | 383      | 383      | 1.000 |
| 106 | 1.5833   | 0.232   | 225     | 380       | 380      | 381      | 0.997 |
| 107 | 1.5792   | 0.218   | 225     | 379       | 379      | 380      | 0.997 |
| 108 | 1.5792   | 0.214   | 225     | 379       | 380      | 380      | 1.000 |
| 109 | 1.5917   | 0.220   | 225     | 382       | 381      | 383      | 0.995 |
| 110 | 1.5917   | 0.220   | 225     | 382       | 383      | 383      | 1.000 |
| 111 | 1.5792   | 0.229   | 225     | 379       | 380      | 380      | 1.000 |
| 112 | 1.5833   | 0.218   | 225     | 380       | 379      | 381      | 0.995 |
| 113 | 1.5792   | 0.248   | 225     | 379       | 379      | 380      | 0.997 |
| 114 | 1.5792   | 0.229   | 223     | 379       | 380      | 380      | 1.000 |
| 115 | 1.5917   | 0.220   | 224     | 382       | 383      | 383      | 1.000 |
| 116 | 1.5917   | 0.233   | 200     | 382       | 382      | 383      | 0.997 |
| 117 | 1.5792   | 0.222   | 225     | 379       | 378      | 380      | 0.995 |
| 118 | 1.5875   | 0.214   | 225     | 381       | 382      | 382      | 1.000 |
| 119 | 1.5792   | 0.218   | 224     | 379       | 380      | 380      | 1.000 |
| 120 | 1.5792   | 0.214   | 225     | 379       | 380      | 380      | 1.000 |

## 4 Sturmian ghosts (critical slope, now a theorem)

Words  $b_n = 1 + \mathbf{1}\{\{n\alpha + \theta\} \geq \alpha\}$ ,  $\alpha = 2 - \log_2 3$ . By the repetition-rigidity theorem these are non-realizable for *every* slope and intercept; the data below shows the same diagonal-riding signature as the candidates.

| $\theta$ | $ D_3 $ | $B_{240}$ | rep bits | mod bits | fill  | rep bits at $N=100/200$ |
|----------|---------|-----------|----------|----------|-------|-------------------------|
| 0.000    | 0.447   | 380       | 381      | 381      | 1.000 | 159/316                 |
| 0.100    | 0.447   | 380       | 380      | 381      | 0.997 | 157/318                 |
| 0.200    | 0.447   | 380       | 377      | 381      | 0.990 | 159/317                 |
| 0.330    | 0.447   | 380       | 381      | 381      | 1.000 | 155/318                 |
| 0.500    | 0.453   | 381       | 379      | 382      | 0.992 | 159/317                 |
| 0.610    | 0.453   | 381       | 376      | 382      | 0.984 | 160/318                 |
| 0.750    | 0.453   | 381       | 382      | 382      | 1.000 | 160/317                 |
| 0.900    | 0.447   | 380       | 381      | 381      | 1.000 | 160/318                 |
| 0.057    | 0.447   | 380       | 381      | 381      | 1.000 | 159/318                 |
| 0.085    | 0.447   | 380       | 381      | 381      | 1.000 | 159/318                 |

Example Sturmian words (first 120 letters):

$\theta = 0.00$ :

122121221212212122121221212122121221212122121221212  
212122121212212122121221212122121221212212122121221

$\theta = 0.10$ :

1221212122121221212212122121221212212122121221212  
1221212212122121221212212121221212212122121221212

$\theta = 0.20$ :

1212212122121221212212122121221212212122121221212  
1221212122121221212212122121221212212122121221212

$\theta = 0.33$ :

1212212121221212212121221212212121221212122121212  
12122121221212212122121221212122121221212212121212

## 5 The negative cycles, computed explicitly

The three known negative cycles anchor the ghost families: each periodic valuation word below has its unique 2-adic realization at a *negative* integer, so the least positive representatives of its realizability classes ride the ceiling  $2^{B_N+1} - |x|$  — the periodic ghost lines of the trajectory zoo. Throughout,  $S(x) = (3x + 1)/2^{v_2(3x+1)}$  and a period- $p$  word  $w$  with sum  $B$  satisfies the affine cycle equation

$$x = \frac{A_w}{2^B - 3^p}, \quad A_w = \sum_{j=0}^{p-1} 3^{p-1-j} 2^{B_j}, \quad B_j = b_1 + \dots + b_j.$$

Negative cycles are exactly those with  $2^B < 3^p$  (denominator negative).

**$x = -1$ : word (1), period 1**

$$3(-1) + 1 = -2 = -2^1 \cdot 1, \quad v_2 = 1, \quad S(-1) = \frac{-2}{2} = -1.$$

Cycle formula:  $p = 1$ ,  $B = 1$ ,  $A_w = 3^0 2^0 = 1$ , and  $x = 1/(2^1 - 3^1) = 1/(-1) = -1$ . This is the 2-adic limit of the all-ones word:  $b \equiv 1$  forces  $x \equiv -1 \pmod{2^k}$  for every  $k$ .

**$x = -5$ : word (1, 2), period 2**

$$\begin{aligned} 3(-5) + 1 &= -14 = -2^1 \cdot 7, & v_2 &= 1, & S(-5) &= -7, \\ 3(-7) + 1 &= -20 = -2^2 \cdot 5, & v_2 &= 2, & S(-7) &= -5. \end{aligned}$$

Cycle formula:  $p = 2$ ,  $B = 1 + 2 = 3$ , partial sums  $B_0 = 0$ ,  $B_1 = 1$ :

$$A_w = 3^1 2^0 + 3^0 2^1 = 3 + 2 = 5, \quad x = \frac{5}{2^3 - 3^2} = \frac{5}{8 - 9} = -5.$$

Letter mean  $\frac{3}{2} < \log_2 3$ : in the positive world this word ascends, the gentle green dotted line of the trajectory zoo.

**$x = -17$ : word (1, 1, 1, 2, 1, 1, 4), period 7**

$$\begin{aligned} 3(-17) + 1 &= -50 = -2^1 \cdot 25, & S: -17 &\mapsto -25, \\ 3(-25) + 1 &= -74 = -2^1 \cdot 37, & -25 &\mapsto -37, \\ 3(-37) + 1 &= -110 = -2^1 \cdot 55, & -37 &\mapsto -55, \\ 3(-55) + 1 &= -164 = -2^2 \cdot 41, & -55 &\mapsto -41, \\ 3(-41) + 1 &= -122 = -2^1 \cdot 61, & -41 &\mapsto -61, \\ 3(-61) + 1 &= -182 = -2^1 \cdot 91, & -61 &\mapsto -91, \\ 3(-91) + 1 &= -272 = -2^4 \cdot 17, & -91 &\mapsto -17. \end{aligned}$$

Cycle formula:  $p = 7$ ,  $B = 1+1+1+2+1+1+4 = 11$ , so  $2^B = 2048 < 2187 = 3^p$ ; partial sums  $(B_0, \dots, B_6) = (0, 1, 2, 3, 5, 6, 7)$ :

$$A_w = 3^6 \cdot 1 + 3^5 \cdot 2 + 3^4 \cdot 4 + 3^3 \cdot 8 + 3^2 \cdot 32 + 3 \cdot 64 + 128 = 729 + 486 + 324 + 216 + 288 + 192 + 128 = 2363,$$

$$x = \frac{2363}{2^{11} - 3^7} = \frac{2363}{2048 - 2187} = \frac{2363}{-139} = -17 \quad (139 \cdot 17 = 2363).$$

Letter mean  $\frac{11}{7} \approx 1.571 < \log_2 3 \approx 1.585$ : again sub-critical, again a positive-world ascender. Note how close  $11/7$  is to  $\log_2 3$  — the  $-17$  cycle is a near-balanced word, the best rational approximant realizable at small height, and exactly the shape that the Baker cycle bound ( $x_{\min} \leq Cp^{\tau+1}$ ) governs.

## 6 The June 2026 constraints, and the surviving candidate list

Since this catalog was first compiled, the program proved a stack of further theorems (companion notes: narrow-band exclusion; two clocks; universal ghost clock; CIC round chain; faithfulness). This section records what they jointly do to the counterexample profile, and then lists what now survives.

### The new constraints, in catalog form

- *Exact narrow-band exclusion.* No acyclic orbit dwells more than  $CY^{\theta(D)}$  consecutive steps in a band  $[Y, 2^{2^D}Y]$ ; the count of confined words is an exact integer (no Perron estimates), and the counting capacity is exhausted at  $D^* = 1.246$  (band ratio 5.63). Bounded-discrepancy words ( $\sup |S_j| < 1.246$ ) are unconditionally non-realizable by divergent orbits — the Sturmian section above is now a corollary. In a ratio- $\leq 2$  band the dwell bound is *logarithmic* (the dyadic strip carries one word per phase).
- *The phase diagram has two arithmetic lines.* Vertical:  $D^* = 1.246$ . Horizontal: the binary bias ceiling  $\varepsilon_{\text{bin}} = 0.449$  — calibrated  $\{1, 2\}$ -words cannot carry more order-3 shadow bias than this; exceeding it requires letters  $b \geq 3$ , which are descent-heavy. Saturation  $\varepsilon^* \approx 0.54$ .
- *Every letter is a metered ghost approach.*  $b=1$  runs meter  $-1$  (length exactly  $v_2(x+1)-1$ );  $b=2$  runs meter the trivial cycle  $+1$  (depth decrement exactly 2, parity handoff);  $b \geq 3$  letters read  $v_2(x + \frac{1}{3})$ . Universally: every signature  $\sigma$  of depth  $K$  has phantom root  $\rho_\sigma = C_\sigma/(2^K - 3^\ell)$  (the cycles above are instances), alignment drops exactly  $K$  per ridden block, rides last exactly  $\lfloor (a-1)/K \rfloor$  blocks, and alignment  $a$  at height  $x$  costs  $2^a \leq |2^K - 3^\ell|x + C_\sigma$ . Amortized over *all* ghost families of depth  $\geq 3$ , the available drift gain is  $R = 0.0882$  bits per letter against a 0.415-bit budget.
- *Supply is consumed without replacement.* An acyclic orbit visits each integer once: each dyadic band holds only  $2^{k-g}$  points of depth  $g$  per ghost, ever. Consequences: at most  $2^{k-3}$  upward crossings of level  $2^k$ ; a divergent orbit spends  $O(X \log X)$  lifetime steps below  $X$ .
- *The survival coin.* Parsed at 1-run starts, life against the letters- $\leq 2$  regime is the reading  $v_2(3^\ell m - 1) = v_2(m - 3^{-\ell})$ , death iff odd. On Haar fibers the coin is exactly i.i.d. fair-thirds; on ghost-aligned seeds it is deterministic (LTE): Mersenne  $2^g - 1$  survives round one iff  $g$  is odd with  $v_2(g-1)$  even; the 3-smooth families  $2^g 3^a \mp 1$  cascade in closed form, with round-two readings such as  $v_2(a' + 2 - \text{Log}_3(-5))$  — the 2-adic *logarithm of the  $-5$  ghost above* — and every tested seed dies by round 7.
- *The  $\{1, 2\}$  alphabet has an entropy deficit.*  $\theta_{12}(D) \leq 0.97907 < 1$  at every band width: for an orbit with no letter  $\geq 3$  the phase diagram's allowed region is empty — no band of any ratio offers refuge.
- *The blind spot bites.* A positive integer has  $\log_2 x_0$  bits, period: after  $\approx \frac{2}{3} \log_2 x_0$  steps the word has consumed them all and every later letter is forced by the  $\times 3$  carry cascade. The infinite word has Kolmogorov complexity  $\leq \log_2 x_0 + O(1)$  while repetition rigidity demands near-maximal factor complexity in banded windows. Realizable words are a countable, computable family (of the  $2^{30}$  depth-30 words over  $\{1, 2\}$ , exactly 66 are realized below  $10^6$ ).

### The composite profile

A counterexample must therefore be: a divergent orbit, fleeing upward (escape  $O(X \log X)$ , single excursions below  $X$  sublinear), never pausing in any narrow band, calibrated to mean  $\log_2 3$  yet carrying non-coboundary bias below the ceilings (0.449 binary,  $\approx 0.54$  saturated), financing every ascent through without-replacement ghost-depth pools and an amortized 0.088-bit census margin, beating an i.i.d. fair-thirds coin forever on its  $+1$ -handoff parities, with a word of near-maximal local diversity generated — after a free opening of  $\frac{2}{3} \log_2 x_0$  moves — by a fixed finite carry transducer with no further choices.

### The candidate list, June 2026

No specific integer candidate exists: every concrete object ever computed in this program is either a 2-adic ghost or dies with a finite certificate. What survives is six structural profiles. For each we list its best known approximant and the meter that taxes it.

| profile                                                                                                                                                               | best known approximants                                                                                                                                                                           | taxing mechanism / status                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| C1. Carry-cascade pseudo-random divergent (generic profile: calibrated, biased, high-complexity)                                                                      | the 120 pilot words of this catalog (all ghosts); integer side: survival champions 27, 4591, 31911, 517191, 687871 (depths 17, 27, 33, 38, 50; 59–76% of each life is in the forced carry regime) | fair-thirds survival coin; census margin $4.7\times$ ; all known members die; champion height grows $\approx 2^{D/2.2}$ with depth |
| C2. Binary violator: letters $\{1, 2\}$ forever ( $\mathcal{C}_\infty \cap \mathbb{N}$ , the CIC class)                                                               | none positive; the 2-adic members are the Sturmian-type ghosts of §3 and their high-complexity cousins                                                                                            | entropy deficit $\theta_{12} < 0.9791$ : no band of any width is a refuge; parity coin must read even forever                      |
| C3. Parity-channel rider (word biased toward $(1, 1, 2)^\infty$ )                                                                                                     | integers 2-adically near $-19/11$ ; least representatives ride the full diagonal (ghosts)                                                                                                         | the $K=4$ clock of $-19/11$ : alignment $a$ costs height $11x + 19 \geq 2^a$ , rides last $\lfloor (a-1)/4 \rfloor$ blocks         |
| C4. Ghost-aligned seeds (3-smooth neighborhoods $2^g 3^a \mp 1$ , incl. Mersenne)                                                                                     | round-one survivors $2^5-1, 2^{13}-1, 2^{17}-1, \dots$ ( $g-1 \equiv 2^{2t} \pmod{2^{2t+1}}$ ); family champion $2^{705}-1$ ( $n=704$ : six rounds, readings 8, 2, 2, 2, 2, 2, 3)                 | deterministic LTE cascade; ghost-log readings; every tested seed ( $n \leq 3000$ ) certified dead by round 7                       |
| C5. High cycle (near-balanced periodic word, $B/p$ a convergent of $\log_2 3$ : $19/12, 65/41, 84/53, 485/306, \dots$ )                                               | the $-1, -5, -17$ cycles above (negative side); no positive candidate below current $k$ -cycle exclusions                                                                                         | cycle formula $x = A_w/(2^B - 3^p) + \text{Baker pinning } x_{\min} \leq Cp^{\tau+1} + \text{narrow-band period localization}$     |
| C6. Exponent-level conspirator (round-chain exponents tracking 2-adic digits of ghost logarithms, e.g. $N_0 = \text{Log}_3(-5) = 2556611305511861515 \pmod{2^{64}}$ ) | none known; alignment at exponent level costs height at exponent level                                                                                                                            | the lifted clock: ghosts-of-ghosts; $p$ -adic Baker bounds the readings effectively                                                |

The list should be read as a specification of where a counterexample would have to hide, not as evidence that one exists: profiles C2–C6 are each pinned by an exact meter with a positivity cap, and C1 — the generic impostor — is the quenched calibration wall itself, the single open core to which every known framework (ours, Tao’s, the LLM-collaboration manuscript audited in the companion notes) independently reduces.

## 7 Collatz Elevator Pitch

One-liner. "We can prove the average number loses this game. We can’t prove that this particular number does — and every number is a particular number."

Thirty seconds. "The rule is simple: take a number — if it’s odd, triple it and add one; if it’s even, halve it. The conjecture says everyone eventually crashes down to 1. Now, the ups and downs of any number look exactly like coin flips, and the coin is rigged downward — so on average, everything falls. That part is basically understood. The problem is that nothing here is actually a coin flip. Each number’s entire future — every up, every down, forever — was fixed the moment you chose it, written into its digits. So to prove the conjecture, you’d have to rule out that somewhere out there is one number whose predetermined script just happens to dodge the crash forever. Probability can’t do that: probability talks about most numbers, never about that one. It’s the same wall as: we can prove almost every number has perfectly shuffled digits, but nobody can prove  $\sqrt{2}$



does — even though  $\sqrt{2}$  is sitting right there, fully determined."

The twist, if she wants one more floor. "Here's what makes it maddening — and what the last year of work actually showed. We've cornered the hypothetical escape artist. It can't follow any pattern: we proved that if its moves ever repeat, even once, in the wrong circumstances, that repetition snaps shut into a loop, and all loops are dead. So a survivor would have to move in a way that's indistinguishable from pure chance — never repeating, perfectly balanced on the knife edge between rising and falling. But it also can't actually be random, because random walkers provably crash. It would have to be a perfect impostor: a fully scripted number flawlessly impersonating chance, forever, while hiding a secret bias it needs to stay alive. We're fairly sure that impostor can't exist — being biased and looking unbiased pull against each other — but proving a specific, predetermined thing can't perfectly imitate randomness is something mathematics doesn't yet know how to do. That's the whole obstruction, in one sentence: we can beat every strategy, but we can't beat 'no strategy.'"

A note on your original phrasing: "deterministic numbers that behave totally randomly" invites the response "then they crash, like random things do —done!" The fix is to put the burden where it really sits: deterministic numbers that we can't prove aren't perfectly disguised as random. The casino version also lands well: "We've proven the house always wins on average. The conjecture asks us to prove no individual gambler beats the house forever — when every gambler's every move was decided before they walked in."

## A The 120 candidate words

Alphabet  $\{1, 2, 4\}$ , length 240, row-wrapped at 60 letters.

```
#1 (mean 1.583,  $|D_3|=0.214$ ):
11141121112112111112114421114112222111122212112122212112221
1122211224112121222212121211111121221222111411112211121
2221111111124222221111112121141121211121422111414212111141
2112114122221211111112222212114121211221411121111114422241

#2 (mean 1.579,  $|D_3|=0.214$ ):
222222222222211111121121142111121142111122122221114241211
1111124121112114221111411121111221122122121111112121111211
2122221112121111422142411121112141121114421211114121114221
21121111111122121111212112211212111411211412422212222111

#3 (mean 1.579,  $|D_3|=0.214$ ):
222112111112112421221112111211112121221112111211221144412
4222221112212121111111211111121141212111222422214111212111
11241111211222421112111211122121122111124112111224111121221
2241221141121111221224121124211222121114411211121212111122

#4 (mean 1.583,  $|D_3|=0.239$ ):
111111111111111111111111111111111111111111112221141142141221114212
11142114111221421242112211441122411112111221212211222221111
111112221221211111111111121421121124112212121142244122121211
121411412211121111211221111422424421212112121142112221121111

#5 (mean 1.592,  $|D_3|=0.220$ ):
1111111111111111111111111111121121111412414211121212411112242
1211211211212111142111412221112421221112112221111111122111
2212221124121121114224121122224421212141111124111221221212
2122112114111112211214211112221112222414212211221214211224

#6 (mean 1.579,  $|D_3|=0.215$ ):
22222224222224211211114124212114112212222111112112241221
1111211211111111111221111222242111421221421111111242221
1111122211111211122214111122221121212421111141112121222
111112111112111122112212142111411221211412141112121212112

#7 (mean 1.592,  $|D_3|=0.215$ ):
12111211121121212111122412111212141121211411241122124222211
2212211111121111122121222421411121112112111222111211112112
2124221411112214124211112222141121421121112121211112211111
1114121222242111111221211221111212421212114112211222141211

#8 (mean 1.579,  $|D_3|=0.216$ ):
12211111111211211111411212221111121111211111211142211414
222221141211111421222112121111112114122121211122121211421121
4121211241122111124221122211111141142121112222411112111221
21412212111411211121122114122411111211211241212122121122221

#9 (mean 1.579,  $|D_3|=0.229$ ):
2222222222242222222222222222222211122211221144111
12112112211122111221111112111421241111222111211211122241111
```

2412121211411111211121111212211112211121111412121214211111  
1111221122111112111111422421211211112211111111112112421

#10 (mean 1.579,  $|D_3|=0.233$ ):  
111112111222121222121211114111414222441214211212124111122211  
111111111121121421111411122221211224211422141122144142111121  
1421411121221111121222241111111111111122422121212211111111  
1112214221112121111111142211121111212111211121114122221

#11 (mean 1.579,  $|D_3|=0.229$ ):  
2222224222224222222112211221241221221112211142211112122112  
1111121112112121111211122112112112211122241121211241141  
12121111421112112112211141212121121422211112112411211222121  
2242212122121121114212111214122111111221111111212211111

#12 (mean 1.579,  $|D_3|=0.216$ ):  
42221121211412112224411122221112411221221112121112441124421  
11222211111211121112212111221221211111111121242212112122  
114122214121111112412411112114112121111141112221141211212  
2111121112111111222421121114211111212212121111212211221

#13 (mean 1.592,  $|D_3|=0.233$ ):  
11111111111111111111111111111111111111111111111222212111121  
12212421411221111112211111222122141114421122211212442114111  
121241421122224122111422121211222421122114421142422121221111  
21112414421212211212121222111211124111211121121114212122141

#14 (mean 1.579,  $|D_3|=0.215$ ):  
242222222422222221111111211221111111111111111221121224211  
2111212122221121221214121121221121221211212111121142111121  
2121112211141111211212441222112111221112212412411211222212  
211211111111212121411411112124111224141124122121111211411

#15 (mean 1.592,  $|D_3|=0.216$ ):  
11111111111111111111111111111111111111111111124411122111211112112211221  
121221121141241214211241221111111224121122222211111222211122  
122212222142121212142222241121411121121112221411414111122111  
211211221121242112114124211211224114212222211141122222111111

#16 (mean 1.583,  $|D_3|=0.214$ ):  
2122141111121111212111112122111212111121122414224112111221122  
2411221421212111111241211244411221221111112221212422211214  
2111214211111121221111242112212114121111122112211111121222  
1141212114112111111241222121221211112121212212121112111211122

#17 (mean 1.592,  $|D_3|=0.215$ ):  
11121121224212122211221222111112122421112112112222114111212  
211111242122412112112121112211141122141121124221112221211411  
4111114212112211121122111122111211111122122112122112114112  
212221111111114111411211214111111222112124211121122242421222

#18 (mean 1.579,  $|D_3|=0.218$ ):  
22222224242211124111122122121141122122112121221122114241112  
11122421122111212111411112121141111222121121212412111121222  
12114121111222221112111122114124141122111221111212212221111  
212214221121121111212121111122211421121112211112111121111

#19 (mean 1.592,  $|D_3|=0.220$ ):  
1124111112111121211124111212111212111114122111112111112221  
121212111212212112121114111111112141122122442121122112222112  
2114142111411121122122441111121121111211222141241114121222  
1111211221244112121111221212411121212124242212112211111111

#20 (mean 1.579,  $|D_3|=0.260$ ):  
222222422222222222222222222222212122111112211111221124112  
21111211221122412111411112221121221211222111111412122212212  
42111421212422121211211221111112112211112211122112212422111211  
121211111121222121111211111111111141121112411212221222121111

#21 (mean 1.579,  $|D_3|=0.272$ ):  
2222222222222222222222222222222121211112212211121122222121  
11111122121121111222111121111211112221112122111141121414121  
111211211121422211211212121122421212211112414212111121121121  
121121112111111221212112211211222122212441221211111412122

#22 (mean 1.592,  $|D_3|=0.220$ ):  
11111111111111111111111111111211112242124211142124122111112241  
221114214212211122221221414421121112221121141222111211112212  
11411111211141121211112111122441121211122112122242111211114  
121111121211212124242212121241212211122411111212122122211212

#23 (mean 1.587,  $|D_3|=0.228$ ):  
142222112411411141112111121121122414422122111111211211222412  
22212111121122121111121112211411212121411111211111221122122  
1242212121111141214211211222111121211114121112421211112112  
1224111111111111111211112222111122424411124114214111122111

[illegible]

14121444211112412422412112121112142121411211122111111241121  
221111141112111211121111111221122114222111111121214112

#39 (mean 1.587,  $|D_3|=0.221$ ):  
224111212221141111122412221141121111111411212111241211211  
111211122221214111422114112211221122121112122121111211111  
24141121112124212121114222211114121121121112121212111111  
1442212111121241221221112221121121114221212111224442112211

#40 (mean 1.579,  $|D_3|=0.218$ ):  
2222222222114112211111121411121111211112111211122241122111  
112121411111224111211112222122112211122112112212111412142212  
2221221222212211211112111411222112121111214111221122211411  
12111121121241141221221221212121141122144121111111242111212

#41 (mean 1.592,  $|D_3|=0.225$ ):  
111111111111111111111111111112211211111122141221412211  
211112221112141221122111122211111122111211212142124142421211  
42212111114111114111124111212121222111222122214122221221212  
2421212241114214422121111222111212411114121224211112412212

#42 (mean 1.579,  $|D_3|=0.238$ ):  
222222222222222222222222222211221114112111111141124121111  
1212411124111221121211111121112112111122211112112212112121  
2111141211111221111221221212121112121112112121111121221111  
21224111241112212221221111221112422211411212121224222212441

#43 (mean 1.579,  $|D_3|=0.215$ ):  
22242224222112122421121412111211124411212111112111124111222  
1111122211122241221111111112121211421121111212112224214221  
22111121211112212211212111121111212211112221221121111122214  
2122412112111122121211112112114221111114211111212221412121

#44 (mean 1.592,  $|D_3|=0.215$ ):  
1111111111111111121112212211221111142111221111224211121211411  
4222111112211111222442111112211112122412111212221412211422211  
112214211221111211112212241121211221221121122121112111121214  
112112111121411112222114122221121212221224411442221222112112

#45 (mean 1.587,  $|D_3|=0.217$ ):  
1211212122421221114141111111112121224112211111111211112111  
12211222211211212224111241141222241242141211221121221421412  
11111112221221211142211112122111211211412121111222222211412  
1111111221212121111121214111241111141121222112142211111141

#46 (mean 1.579,  $|D_3|=0.215$ ):  
22224222222222222222222121111121111211214221114411221111112  
1241111111121211111211122221111212212112111122111224122114  
2111214441111142122111222211112111112122121121112222211111  
21411214124121411211421111212121112114211211112111121212

#47 (mean 1.579,  $|D_3|=0.229$ ):  
24242222222222222222222422442222221221111241111441142111111  
1111212112111112414121111212212121112221121211211222111211  
114111212111221111122121222211221121212111111112122211111  
121421121122212242111222212221212211111111112111212121411

#48 (mean 1.579,  $|D_3|=0.215$ ):  
242222224212211121111111121211124411411111122422121421112  
211121222212221241112112212212212121212121242111141111211  
1214141111111112111411211212222142121112122112122212111221  
242421221211111111111222211122211211141121111121121212112

#49 (mean 1.592,  $|D_3|=0.215$ ):  
111111111141121142222122114411211221421214242121121244211111  
212221121121121212142122114112121112122121211222221212111211  
221111211221111111121121241112111221111221212111212111122  
2222111121141212222114421421222111112411112222412121111111

#50 (mean 1.592,  $|D_3|=0.216$ ):  
12211212142111124212112211212121122121222142142111112111121  
1412211111414211112221221214121221111144112111111111121221  
21114412111211121121212211221122221214121211111121114111212  
121411242124212212122122124112211114121112112122421111221211

#51 (mean 1.579,  $|D_3|=0.229$ ):  
222222222122124211122412221122112112111221122214122111112112  
21112441121112142111211141111221111222442111111121122211121  
21212212111112222211122121141221222112122111211211122112221  
11111212222114111411411212111112141122212112122112211121121

#52 (mean 1.579,  $|D_3|=0.248$ ):  
222422222222222222222222222222112111211122111112121  
2111222111111211124111112222112111111222111121121122111211  
24211111112211212242122211244211241412111122111221112111111  
41111221221112121111124112211211212114411211121121121221121

[illegible]

21411121411122122111214111211112121121221224121411122211121  
412111144211212111212412211212122212111212122211222111211222

#68 (mean 1.579,  $|D_3|=0.216$ ):  
222222222222424242411121211112124122114421211211411111221  
21211212111111244212112212222211212111211121211211141211112  
1121211141221211114121112111111221221112211111112112411212  
12212111114111111441122111111122421211212411122211112111

#69 (mean 1.592,  $|D_3|=0.216$ ):  
11111112122411112212111141111122121121224112221222221142111  
111114221211242212141122224242211214242111221111221211222111  
14111111212122112211111212211122114111112422121112111111  
12124112221121411111411122421121112111412121121111421212

#70 (mean 1.592,  $|D_3|=0.216$ ):  
2111421211224122241112241121121121211211211111412112222  
112211121214121221212111111224112121121212111122111214224  
1122211212112121211422112221121121141114222111222411111111  
12412212111412111122122112111214421111144224121111111112

#71 (mean 1.592,  $|D_3|=0.214$ ):  
11111111111111112211214121421112122112211114122144221122222  
1144144124242112221411121121212111112111421222112221212211  
1222242112212111212221241111141121122112111212111121111222  
2112122112142121211112111221121241211211221412121112111112

#72 (mean 1.579,  $|D_3|=0.214$ ):  
24224222221221111141112111241121121212211221111221212121  
22441111422112211122411111121212111211212114212121141242111  
2121112112221122214122411112111121221422121211212111122121  
122212212121211211112111214112111211122411111112111411114

#73 (mean 1.587,  $|D_3|=0.221$ ):  
121141111122411121121124221121212211411222111121111124214211  
12222111111122121112121412111441112212111121211111221212121  
112211111112412421211222122111114121124111111121112112121241  
211122242214111122411211211121222141212122241111221111411241

#74 (mean 1.592,  $|D_3|=0.215$ ):  
111111111111111112122112121422222114111122121212121122111  
111221221141242112222111122122121211212224121122221142212212  
212221221112121111122242122111111121212121211112114244142  
2214112111241112114111211111122111141224221222211411211124

#75 (mean 1.579,  $|D_3|=0.215$ ):  
221124111421212121211122221111112421211212121121141221211  
2121114121141111212111121141121241211241111211221221224412  
22111122221111212212211111121212122221141112111422221111211  
11421122112112121221211121221241241121211112122112111114121

#76 (mean 1.579,  $|D_3|=0.225$ ):  
2222421112214111221111124211111121112112124111111214111112  
42111112111121212111112241211121421212111111112221122122  
21112212122121114112111242212421112411221121111111112112414  
211141212112111112111121222121414221111211221114412214421112

#77 (mean 1.583,  $|D_3|=0.241$ ):  
11111112222111211111211112221211414122242142222221242142  
22112211412422121212211211124121112121211121111121222221  
1211221121211422221114222211124111121121112211111221221122  
1212212211242121112121122122121112212211122121111221212112

#78 (mean 1.592,  $|D_3|=0.215$ ):  
111111111111211121222114422112222122222111112112212111412  
112221122212114241411112221211112111411121244111112112222111  
22122212112121112112221441221111111412122221111121222212222  
1111224121112211121111211211142212221421411122111114411211

#79 (mean 1.587,  $|D_3|=0.221$ ):  
141211221111114212212121112121122221112111111114111111124  
112114142221221111211221421222212111211221111114414124222  
11111124112141112111221111124211412111142111221211122112112  
124112141112221112212122112111241141112121221221121111222242

#80 (mean 1.587,  $|D_3|=0.217$ ):  
111112144211112211141212112111211222111122121211121221111121  
21122111112141421221211142214221211121121112112211212122142  
121112212212122211412222121211121111122124111114211122242  
12211224222111241114212121211121242421212111221112111121112

#81 (mean 1.592,  $|D_3|=0.225$ ):  
111111111111111111241411112222112141212121111412114211421  
1122121111112112112124121111122424111111221142112111212141  
22221121112114221112224111242221111114122112122124122211112  
11111212111222121122411421112422124212221421111112141221121

#82 (mean 1.583,  $|D_3|=0.214$ ):  
21211212424121111214112121121221121121211212422111111121214  
224111211121124412221122122111222112121122141142214112424421  
1242112211121111121122241111212121211122111211112112212121  
1111111122122111214112211121111222111122112241211221221111

#83 (mean 1.592,  $|D_3|=0.215$ ):  
112212111124212112211211112112111141122121212221111111122  
22111122411122221222122212111121222112112111422421441221212  
241112221221211121414122442221112241111211212112211112111421  
411112121212114141111122121221411211212111112221112121111

#84 (mean 1.592,  $|D_3|=0.218$ ):  
122111111121141121122212142221142212112221111212122121211211  
121211112112122421411111141121242211112122221111211144112242  
121111111122121122122221221144122111112122212211411111214  
21112121422114211121211121112114112111122212112222221112221

#85 (mean 1.579,  $|D_3|=0.222$ ):  
2121224411222214222122121211112111121114141141211111122  
1221211111211212212121121121221211412221211211121122221211  
2221242122112111211211241212141111221241112211221211112212  
4412212212121111221212241112121212122211211412121211111

#86 (mean 1.579,  $|D_3|=0.238$ ):  
222222222422242222222222211221111212241212211121242112214  
112111111111112112214211111221112111112121121412221111142  
211221111112211112111212111214122121212111122121221122112  
212122221114114242122112121121111222221212112111411112112

#87 (mean 1.579,  $|D_3|=0.229$ ):  
4222222222222224222422214222211122212121112111211122111412  
11122111211111422111221111121124112111221224111111121111  
1121222111111111111114111224112122121111112211211242222  
2211222211111211121124142112122112212224221111212121211221

#88 (mean 1.587,  $|D_3|=0.227$ ):  
2421412121212121214121114111222121211111212211212121212222  
121112221111211111212412111211122122111111121111211212121  
12124212241221211121212221111122121222211212421212121221122  
11224211112222121244222111121414121111121142211111124122

#89 (mean 1.579,  $|D_3|=0.218$ ):  
42222222221141112112211121212122111111212111111122122121  
121122211212122122211124121121114112114211121121211121114  
241221111122111411112241411212111211112422212222221212242  
211221111211212122111111222112111122411142141112412221112

#90 (mean 1.592,  $|D_3|=0.216$ ):  
1111111212121114212212121112122112211121211112142211142111  
21112211112441211122221421211111112121121121142211112421112  
1111212122122221212241421122111212121212121112111411421211  
1111211411221212111122221221411411112244142112111241224111

#91 (mean 1.579,  $|D_3|=0.216$ ):  
42222422221214111212112121214111211114111212111111211112  
1121121212212411211121412212121111111112112224111211122212  
1244122141421144121211211111241221112111122221221212111211  
221221111124122111111121112211211112212114121114214121222

#92 (mean 1.583,  $|D_3|=0.225$ ):  
1111112122211122122114112112211224212222111124221211112112  
124111222411211241121122112111111121224122422111111112212  
21112111221212111211112212111212122242222222211122121212  
11211211222124124242114121111211111221111121242111241211221

#93 (mean 1.592,  $|D_3|=0.220$ ):  
111111111111111111122124212111114141212221211112112212111  
11221411112122111111222124121121111241211212121212122112122  
44242112211142112411212111221121112112114111244114111124122  
221112111211222122121211424212242121411212111411122112122111

#94 (mean 1.579,  $|D_3|=0.214$ ):  
4422222222222242412111111111111211122111121124142112211211  
112112122121224212122411211112211122124212112111411121111211  
141242112111121221111224114121111124122111111111142111112  
22211122111112121112142111212221221112112111121412212112211

#95 (mean 1.579,  $|D_3|=0.238$ ):  
22222222222222242222222222222222222221121111111112212121  
24111111211111211212211124112112111112142121121121211121221  
111241222211114111221221111142224121121111212112111111112  
2111122122211141112212221111242224111111112111222412211111

#96 (mean 1.579,  $|D_3|=0.238$ ):  
222222242222222222212112242111121111212112124121211122122  
2211222212111121114122111111121411221112211212122111221112

12111412211411112211121221111222112124141211212124221211141  
12221211111121124122111112221212111111221121112421122121

#97 (mean 1.592,  $|D_3|=0.233$ ):  
11111111111111111211414222111121221112222111112111142412  
2112114214211142121224111422124111212421114221122212211112  
1122112111111211212122211121114121212222141214111121221  
121411112221411111222112211141121141211241121224411112411

#98 (mean 1.579,  $|D_3|=0.229$ ):  
22222222222222222211141211111221112212221111112121221  
1212111422112111221111412221221111221112111121112121111122  
111212241411111141222221211112111121211212222114111124  
124121222411211211112422412212114211211122111224112111212212

#99 (mean 1.592,  $|D_3|=0.218$ ):  
11111111111111114112211414242242111124124111122112112212  
1221114141121112211112412212114112111122112124422221224221  
12221212221212111111211222224214122211221112221122111212212  
1221112111122221212221112211121112222141111121121212211

#100 (mean 1.579,  $|D_3|=0.215$ ):  
2222222111441121124221121114121221122212141112121111121221  
21411221212221222122212222111212142112212211124411411211111  
114411114112111411411111421122112211211211121212121212211  
2121121111121212212111112221122212211121112112121411211

#101 (mean 1.579,  $|D_3|=0.218$ ):  
222222222222111121212111121121114212112121111221141241211  
12111411111211212211111121212411222211222112121111211122111  
142222112112112111411211121212421222211142422212211421221  
222411211141214121211112111121111212121121111424122111

#102 (mean 1.579,  $|D_3|=0.218$ ):  
222222222222222222222422222222121211224211222122412112  
212121111212121112212211411211114112421111112111121114111  
121122211112111112111114211211142111111111224111244421121  
1122111111121112112141111121111242411221111122112121121121

#103 (mean 1.579,  $|D_3|=0.229$ ):  
22222222221211111211121112241122221121211122122111221111122  
1221111212421121121212111212412121121222442121112121442211  
21121212111111242212112112121124221111221111122211112121112  
12141211121121212211222111241222124121114211111121241121112

#104 (mean 1.587,  $|D_3|=0.217$ ):  
1211422221212121212111421112211112212211222141211212122221  
1111214112111211211221141111122212111122111112111122122222  
2114212112111112111214111421411212221122111121421241221  
21111121412221211412121122211211114221141122121221224211141

#105 (mean 1.592,  $|D_3|=0.229$ ):  
11111111111111111111111111111212122111112122112211211411221  
1121211111111211221112212224211121421124212221121222422222  
11242121211112122211224411122222421222112111222124221222241  
2221122411211241112212121212421122211121121112121241122212

#106 (mean 1.583,  $|D_3|=0.232$ ):  
11111211422221221212111241111121122224114221122122111422111  
21112222121112211222122211211211122221211121212121242221122  
2121221111121122111122121112221114224211214221122112412142  
1221411141212121212221111121111122421112212111122111221112

#107 (mean 1.579,  $|D_3|=0.218$ ):  
242222222221221111111122214121222111112211221111111111111  
121211111114212112211111422224112121111211111111121212221  
222121222111121112121142211112121211412142112111221212212211  
11211111144122122111142222121412222111242221141222211111424

#108 (mean 1.579,  $|D_3|=0.214$ ):  
242222222222212122121122111212112212211211112224214414122  
111211111121111411211112221111112214211112211111142211122  
111111222222221222114141111414211121211111112112121411212  
1112214144111222111121211121111124211222211111211211141

#109 (mean 1.592,  $|D_3|=0.220$ ):  
1111111111112242124214214112211121421222121122111242111214  
21112121111122141141421211121242111111111221221111242142121  
11214221222211112111112412422111111212121111212211211212  
4121111112211112111211111121211142422211412242122221112122

#110 (mean 1.592,  $|D_3|=0.220$ ):  
11111111112221421121222122121111141121421112412411211142121  
211111211111111124112121211112222221241121111422221221122  
2422222111112114212111111121112112244211121111121212124211  
11221221111211111214111221421142122122124211421214112111214



#111 (mean 1.579,  $|D_3|=0.229$ ):  
222222212122121121111212121121111112114212221221111141212  
12411212122121111122121221124111112221221112112111211222111  
1224141111111221124441212212121111121122421222112122112221  
11212122111221121211212124112112211121242111211221422421

#112 (mean 1.583,  $|D_3|=0.218$ ):  
111111111121211121141221221412211122211111211141111122111  
24112142121111212121241142211242211112111222144141214222121  
112121121222111222112111211211111112122111112112111121121  
212442111411122212122112221112442422111211144221211111

#113 (mean 1.579,  $|D_3|=0.248$ ):  
222242222222222222112122211122221112111142211121212111241  
111112212121212242111121122211221121122121211121122111111  
11222211221111212122211211114221142121211214141112211112  
2124112112111121122112112214122221121422212121422211111112

#114 (mean 1.579,  $|D_3|=0.229$ ):  
2222222222222222211111211211121212124111122411112222  
11222221221111121111211211141122211412411111222112411211  
2112142121111111111441211222122111121211421241121211221222  
142212121122221212221121211212114111211211211112214211

#115 (mean 1.592,  $|D_3|=0.220$ ):  
1111111111111111121121121114222221221222242212211211114111  
2222121211121111111114411142212121214111242214124211212112  
121211411121221222111111241214422111222122121114121121241211  
1222114411211212112111211121111211142111122241142221211111

#116 (mean 1.592,  $|D_3|=0.233$ ):  
111111111111111111111111111111111111112112112114212221244  
112111422124111121241222121142222121242122112121222111222212  
221414412114111111212124211412142141212421221111121112222  
11211111112114121121112411111224222141121222124124111121111

#117 (mean 1.579,  $|D_3|=0.222$ ):  
2222222112222111141111122214211221212422112211111122241221  
2112112112112221112111212111211112211222122421111121211422  
111111121111212222411111211211221111211121421111221121222  
41222211221214111112141222221111212114124211241222114112221

#118 (mean 1.587,  $|D_3|=0.214$ ):  
12412224222141112121111121212112122111212111211121422111  
2112211111211111221141111222111122212111211121111242212111  
11122112421112211114112111212211112142112242211122221224144  
12422122221111142121112111124241121222221111121121221142

#119 (mean 1.579,  $|D_3|=0.218$ ):  
2222422222222222222222222222242112112112141111111221142  
11121211121112121222122112111221211122111211211111121124111  
1112111221114211111121111124142122422141112222121111122  
141121122111111121121141211221111121121111141422241211

#120 (mean 1.579,  $|D_3|=0.214$ ):  
22222422222422222212241121212211112111111121211211212111  
21221112112111211111121141221122111211221112224141112111  
12112212121122114121121211412211114111221211111211422444  
4112111112121122241141221241241112121121211111121121211221